Lower bounds for $\binom{2n}{n}$ [duplicate]

Asked 4 years, 6 months ago Modified 4 years, 6 months ago Viewed 2k times

2

3

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What are common lower bounds for $\binom{2n}{n}$?

Edit: I made a mistake in my original question.

It doesn't change my question but there is no reason for me to include the mistake.

inequality factorial gamma-function upper-lower-bounds

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edited Mar 20, 2018 at 19:36

asked Mar 20, 2018 at 19:31



Larry Freeman

10.1k 4 19 54

You could try stirling's approximation on $\frac{(2n)!}{n!^2}$ – vrugtehagel Mar 20, 2018 at 19:34

Stirling's gives roughly 4^n I do not recall the details, but easy enough to do again: – Will Jagy Mar 20, 2018 at 19:34

1 $2n/n = 2$, not n. – John Brevik Mar 20, 2018 at 19:34

1 Here is a tight lower bound for Catalan numbers, which are $\frac{1}{n+1}\binom{2n}{n}$. – mechanodroid Mar 20, 2018 at 19:35

3 Answers

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6

Using the bounds from this answer, we have

$$\frac{4^n}{\sqrt{\pi\left(n+\frac{1}{3}\right)}} \leq \binom{2n}{n} \leq \frac{4^n}{\sqrt{\pi\left(n+\frac{1}{4}\right)}}$$

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answered Mar 20, 2018 at 19:42



robjohn ♦
330k 34 431 806

good; from Stirling's, the squared denominator keeps an extra $\sqrt{n}$ compared with the denominator. I wasn't sure yet – Will Jagy Mar 20, 2018 at 19:44

@robjohn is this a better estimator than Stirling's? – user Mar 20, 2018 at 20:01

1 @gimusi: you can easily notice that $\frac{1}{\sqrt{\pi}} > \frac{2\sqrt{\pi}}{e^2}$, hence robjohn's bound is sharper than yours. Also because the constant in robjohn's answer is optimal. – Jack D'Aurizio Mar 20, 2018 at 20:07

@JackD'Aurizio oh yes of course! Indeed it is a very narrow bound. I will take a look to the derivation. Thanks – user Mar 20, 2018 at 20:09

1 @user: note that $\lim_{n\to\infty}\left[\frac{16^n}{\pi\binom{2n}{n}^2}-n\right]=\frac{1}{4}$, so the $\frac{1}{4}$ in the upper bound is optimal. The $\frac{1}{3}$ in the lower bound can be replaced by $\frac{1}{\pi}$ and the inequality will still be valid for $n\ge 0$. – robjohn ♦ Mar 20, 2018 at 21:29

|

4

Using Stirling's bound

$$\sqrt{2\pi}\,n^{n+\frac{1}{2}}e^{-n}\le n!\le e\,n^{n+\frac{1}{2}}e^{-n}$$

we obtain

$$\binom{2n}{n}=\frac{(2n)!}{n!^2}\ge\frac{\sqrt{2\pi}\,(2n)^{2n+\frac{1}{2}}e^{-2n}}{e^2\,n^{2n+1}e^{-2n}}=\frac{\sqrt{2\pi}\,2^{2n}2^{\frac{1}{2}}n^{2n+\frac{1}{2}}}{e^2\,n^{2n+1}}=\frac{2\sqrt{\pi}}{e^2}\frac{4^n}{\sqrt{n}}$$

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edited Mar 20, 2018 at 19:54

answered Mar 20, 2018 at 19:35



user
142k 12 71 134

1

Erdös had a really nice lower bound, which he used in his proof of Bertrand's Postulate: $4^n/2n\leqslant\binom{2n}{n}$. This follows because

$$(1+1)^{2n}=\sum_{k=0}^{2n}\binom{2n}{k}<1+2n\binom{2n}{n}$$

and then $4^n\leqslant 2n\binom{2n}{n}$. In fact the bound can be strengthened without too much difficulty to give $4^n/n\leqslant\binom{2n}{n}$.

Edit: The stronger bound only holds for $n\geqslant 4$.

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edited Mar 20, 2018 at 21:11

answered Mar 20, 2018 at 19:42



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2,131 9 12